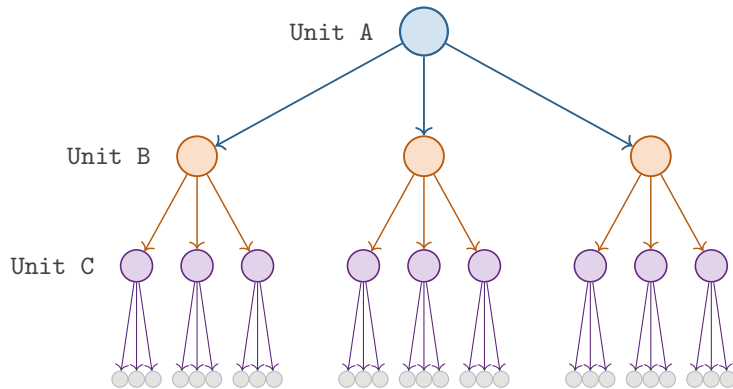


Joint action space

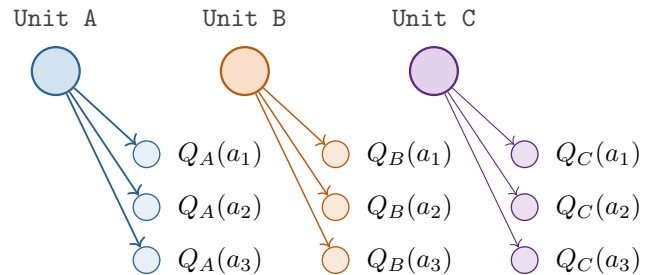
$$k^n = 3^3 = 27 \text{ combinations}$$



27 joint actions to evaluate

CMAB factorization

$$n \times k = 3 \times 3 = 9 \text{ evaluations}$$



$$Q(s, \mathbf{a}) \approx Q_A(a_A) + Q_B(a_B) + Q_C(a_C)$$

9 values to estimate